

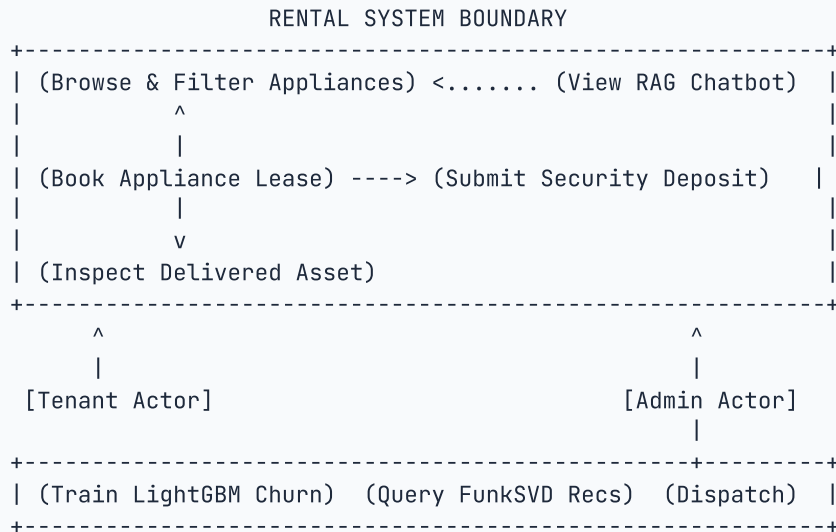
Unified Modeling Language (UML) Specification

Comprehensive Structural & Behavioral UML Diagrams for the Appliance Rental Platform

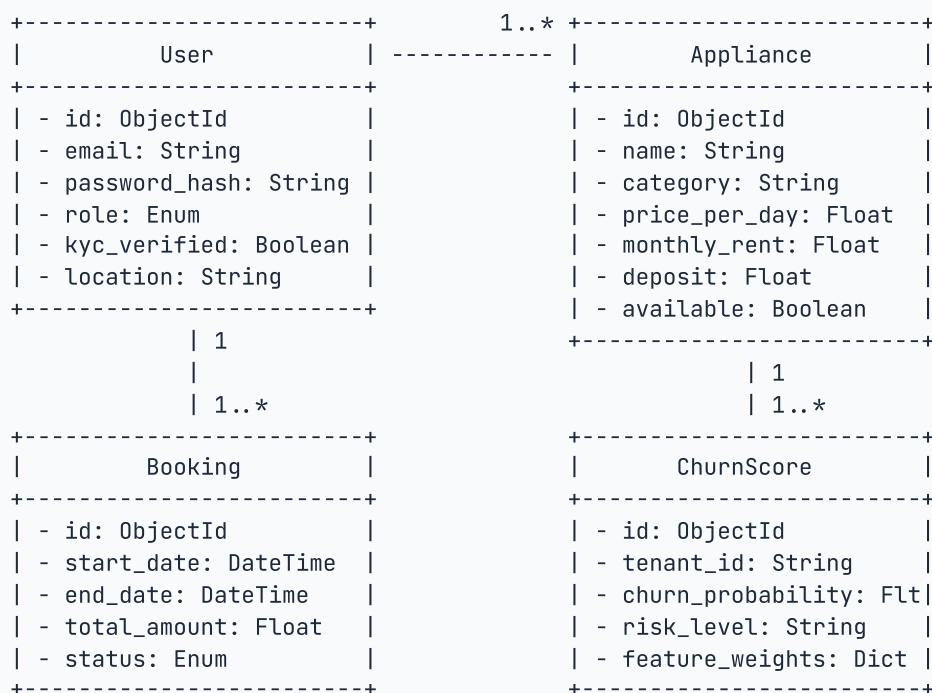
Production documentation artifact for the AI-Powered Household Appliance Rental System (RentAI / SDC2), encompassing enterprise Pure NoSQL architecture, customer churn prevention, multi-paradigm collaborative filtering, and local LLM context assistance.

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Lead Architecture:	Deep Learning & Full-Stack Systems Team	Publication Date:	September 2026
Status:	VERIFIED & APPROVED	Target Platform:	Ubuntu 24.04 LTS / Windows 11 Enterprise

1. Use Case Model



2. Class Diagram (MongoEngine ODM Document Architecture)



3. Booking State Machine Transition Model

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[*] → [PENDING] : Tenant places booking
[PENDING] → [CONFIRMED] : Payment & KYC verified
[CONFIRMED] → [DISPATCHED] : Logistics assigned
[DISPATCHED] → [INSTALLED] : Engineer verifies setup
[INSTALLED] → [ACTIVE] : Lease term commences
[ACTIVE] → [RETURN_REQUESTED] : Tenant initiates return
[RETURN_REQUESTED] → [INSPECTED] : Quality audit passed
[INSPECTED] → [COMPLETED] : Deposit refunded & closed
[PENDING] → [CANCELLED] : Tenant aborts
[COMPLETED] → [*]
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